

Daily Review

Booklet p.10-11 — Getting Around in Space

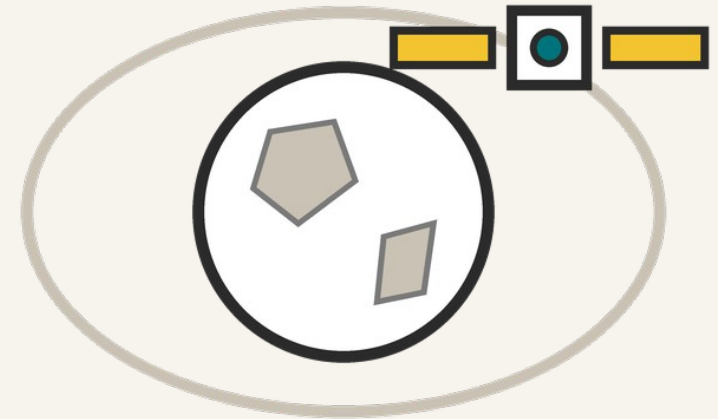
- 1 Engine off in deep space. Describe the motion.
- 2 Describe the catapult manoeuvre.
- 3 How does an ion drive reach a high speed?
- 4 Why is there no friction in deep space?
- 5 What is thrust?

SPACE

Satellites

Weather, GPS and phone calls — all from objects falling round the Earth.

National 5 Physics (S3)



THE BIG QUESTION

One satellite hangs still above this coast. Another crosses it in ninety minutes. What sets the difference?



Scotland from orbit — NASA

PART 1

What satellites do

What satellites do for us, and why they do not fall down.

By the end of this part...

- 1 I can describe the benefits of satellites
- 2 I can explain how a satellite stays in orbit

Show me — Describe each word

SHOW ME BOARDS

1 satellite

2 orbit

3 period

4 altitude

5 geostationary

Task 1 – what are satellites for?

Booklet p.12

Write down as many uses and benefits
of satellites as you can.

Booklet p.12, Task 1

Task 1 – Answers

ANSWERS

Booklet p.12



GPS



Weather forecasts

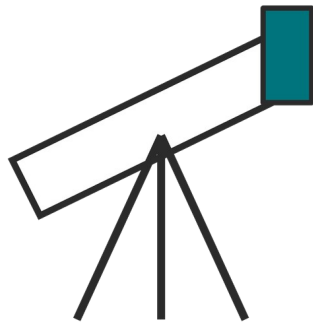


Communications

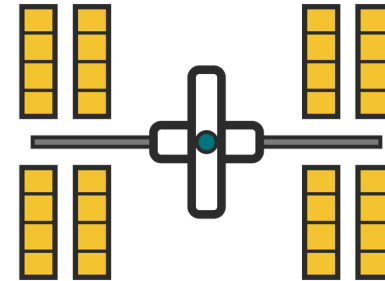
Task 1 – Answers

ANSWERS

Booklet p.12



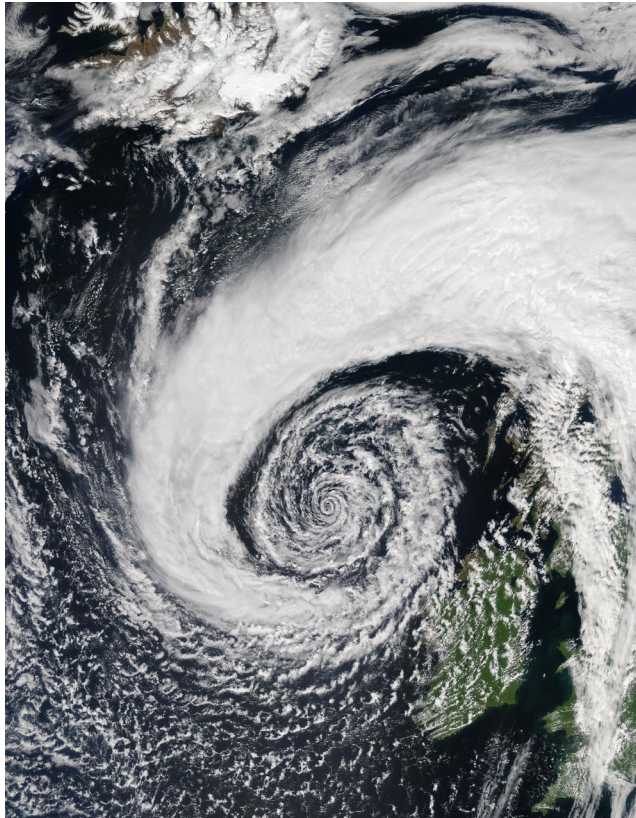
Scientific discovery



Space exploration

Watching the weather

A low pressure system off Ireland



- Satellites photograph the cloud from above.
- This spiral is a deep area of low pressure.
- Forecasters track it as it moves.
- Warnings can come days ahead.

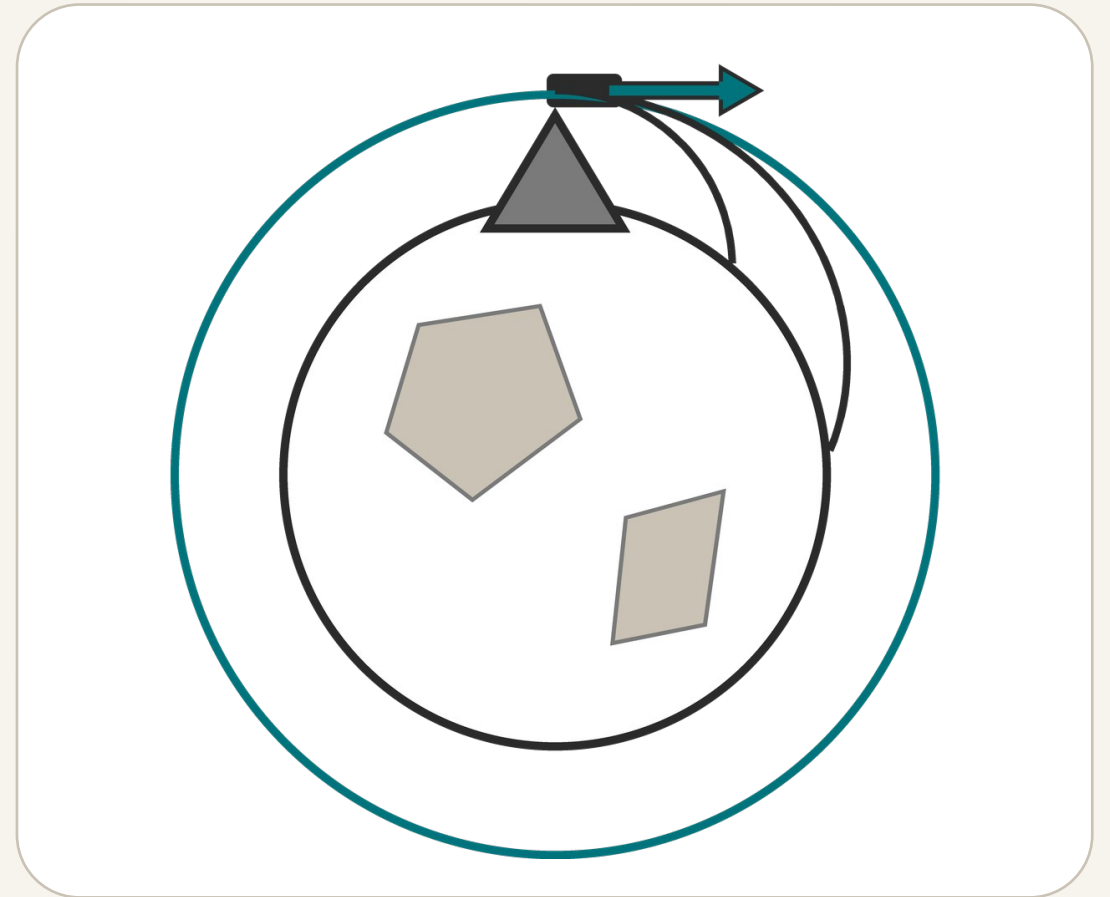


The Whirlpool galaxy — NASA / ESA Hubble Space Telescope

Newton's cannon

Booklet p.12, Task 2

- Fire a cannonball sideways from a very high mountain.
- The faster it goes, the further round it lands.
- Fast enough, the Earth curves away as fast as it falls: an orbit.



Task 2 – Newton's cannon

Booklet p.12

Watch the animation, complete the diagram,
then describe what happens underneath.

Booklet p.12, Task 2

Hinge question

FINGER VOTE

Think first — then show A to E on your fingers, 1 to 5

A satellite orbits 400 km above the Earth.
Which statement about the satellite is correct?

- A** There is no gravity acting on it
- B** It is falling, and keeps missing the Earth
- C** Its engine keeps it up there
- D** It is beyond the Earth's gravity
- E** It moves in a straight line.

Show me — orbits

SHOW ME BOARDS

EXTRA PRACTICE

- 1 What does “period” mean for a satellite?
- 2 What is meant by “altitude”?
- 3 Why does a satellite not crash into the Earth?
- 4 Name a satellite used for scientific discovery.
- 5 How do satellites help forecasters?

Daily Review

Booklet p.12 — Part 1

- 1 Name three uses of satellites.
- 2 Why does a satellite stay in orbit?
- 3 What does the period of a satellite mean?
- 4 What is altitude?
- 5 What is a satellite?

PART 2

Orbits, altitude and period

Choose the height and you have chosen the time for one orbit.

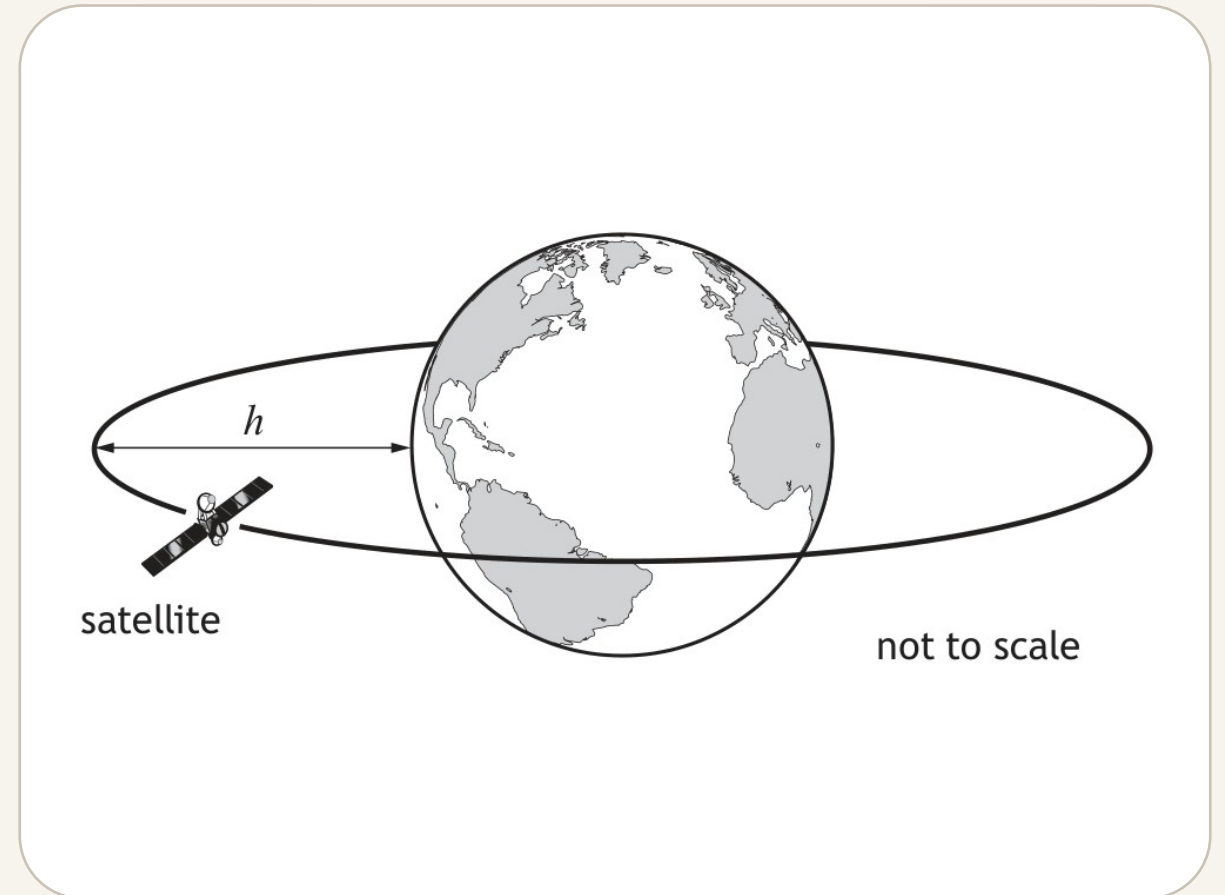
By the end of this part...

- 1 I can state the altitude and period of a geostationary satellite
- 2 I can describe how altitude affects the period of a satellite

Geostationary satellites

Booklet p.12, Task 3

- A period of 24 hours.
- An altitude of 36 000 km.
- It stays above one point on the Earth's surface.



Task 3 – geostationary satellites

Booklet p.12

Complete the definition.

Booklet p.12, Task 3

Task 3 – Answers

ANSWERS

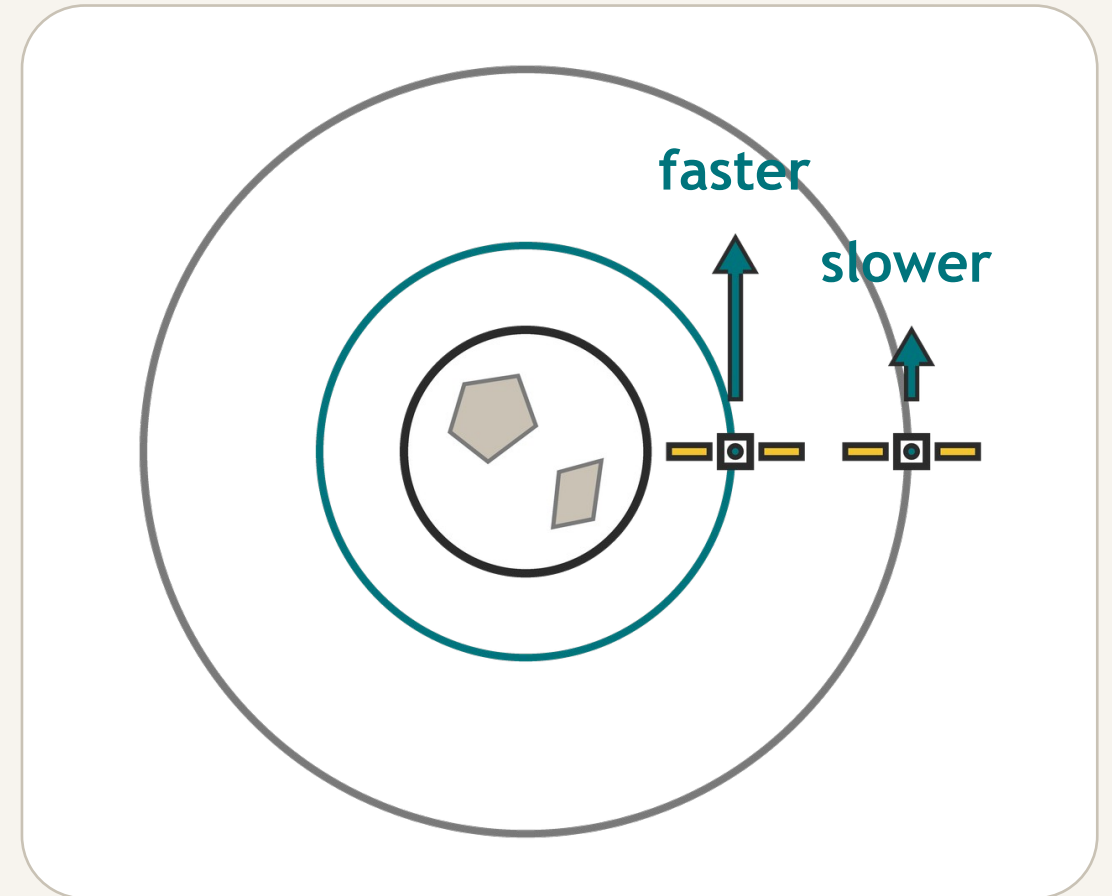
Booklet p.12

Geostationary satellite	
Orbital period	24 hours
Orbital altitude	36 000 km

Why a higher orbit takes longer

Booklet p.12, Task 4

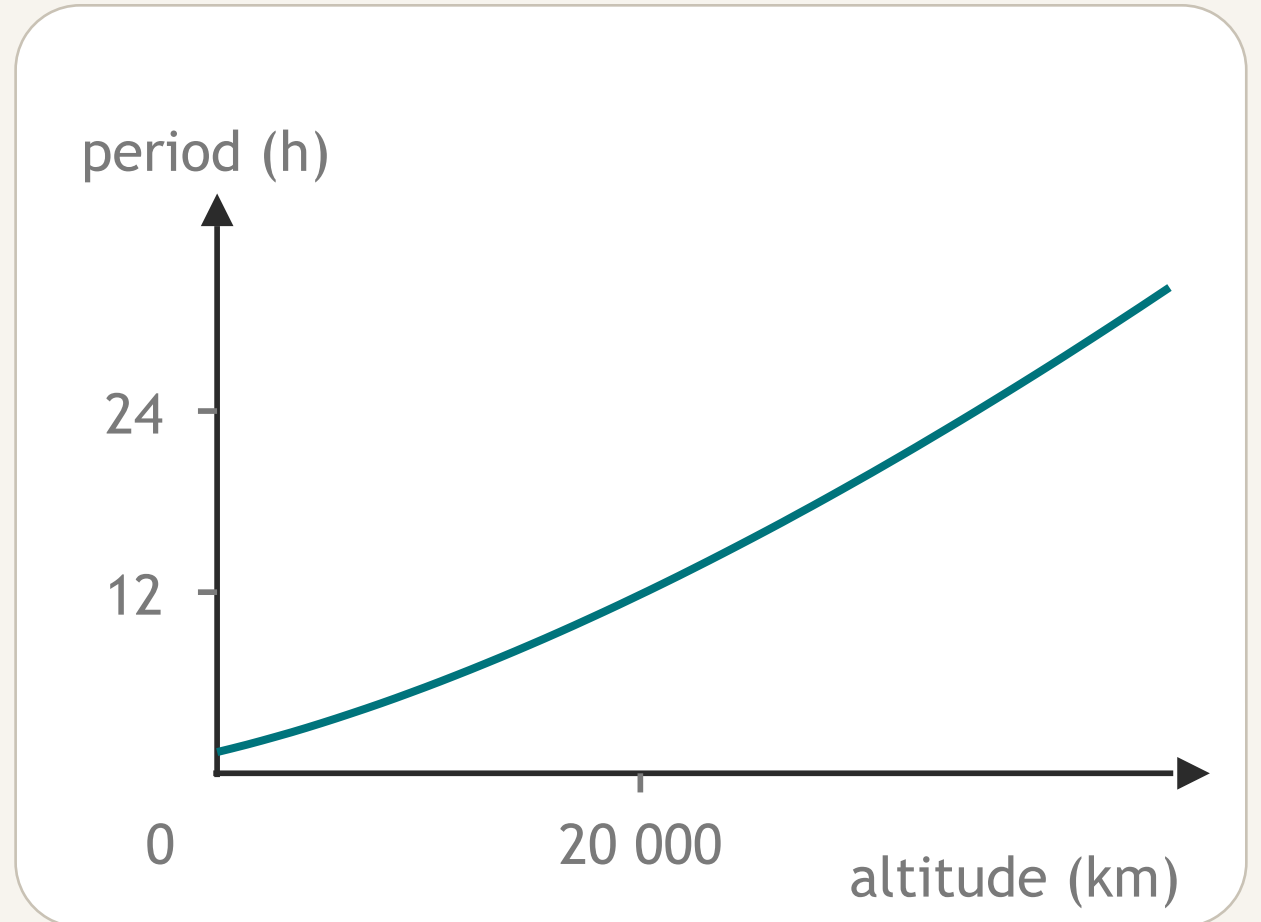
- A higher orbit is a bigger circle: further to travel.
- Gravity is weaker up there, so the satellite moves more slowly.
- Further to go, and going slower — so the period is longer.



Higher orbit, longer period

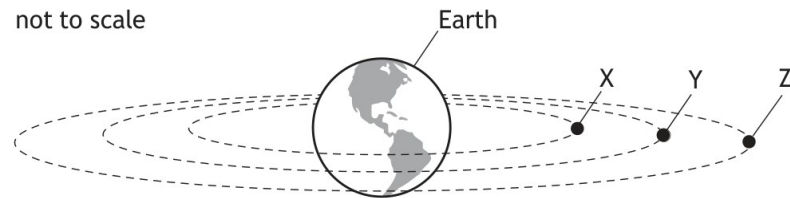
Booklet p.12, Task 4

- The higher the orbit, the longer the period.
- The ISS, about 400 km up: 93 minutes.
- Geostationary, 36 000 km up: 24 hours.



Show me – X, Y and Z

SHOW ME BOARDS



- 1 Which satellite has the shortest period?
- 2 Which has the longest period?
- 3 Z is geostationary. What is its period?
- 4 Can X have a period of 24 hours? Why?

Task 4 – altitude and period

Booklet p.12

Fill in the blanks.

Booklet p.12, Task 4

Task 4 – Answers

ANSWERS

Booklet p.12

- As the orbital altitude of a satellite increases, the orbital period increases.

Show me — predict

SHOW ME BOARDS

Altitude (km)	Period
700	99 minutes
2000	127 minutes
10 000	5.8 hours

1

Predict the period at 1000 km.

2

Is 20 000 km more than 5.8 hours?

3

111 minutes: what altitude?

4

What pattern does it show?

Show me — predict

SHOW ME BOARDS

EXTRA PRACTICE

Altitude (km)	Period
408	93 minutes
1000	105 minutes
36 000	24 hours

1

Estimate the period at 600 km.

2

Is 20 000 km more or less than 24 h?

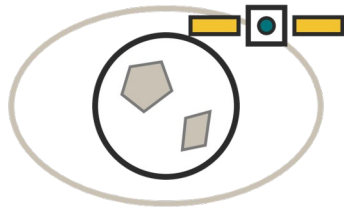
3

Is the satellite at 408 km geostationary?

4

Which satellite goes round most often?

Scotland in space



UKube-1, Glasgow



SaxaVord, Shetland

- UKube-1, Scotland's first satellite, was built in Glasgow (2014).
- SaxaVord on Unst, Shetland, is licensed as a UK spaceport.

Fly the orbits yourself

TRY THESE

NASA Eyes on the Solar System: fly to any planet.
PhET Gravity and Orbits: change the speed and watch.
Try Newton's cannon: how fast is fast enough?

Extension – BBC Bitesize

Booklet p.13

Revise space exploration on
BBC Bitesize.

Booklet p.13, Extension



BBC Bitesize

Checkpoint

- 1 Give three benefits of satellites.
- 2 State the period and altitude of a geostationary satellite.
- 3 Satellite A orbits higher than B. Which has the longer period?

Exam practice

Past paper and specimen paper

- National 5 Physics 2023, Section 2, question 3(a). 2 marks.
- Question 3(c). 1 mark. Predicting from a satellite table.
- Specimen paper, Section 2, Q2(d)(ii). 2 marks. Why the ISS stays up.
- Specimen paper, Section 2, Q8(a). 3 marks. A GPS signal's distance.
- Then booklet p.14, Task 5.

LOOKING AHEAD

Forces on Rockets

Thrust, weight and Newton's laws — what it takes to leave the pad.

Plenary

- 1 The big question: why does one satellite hang still over Scotland?
- 2 Name three things satellites are used for.
- 3 What is the period of a geostationary satellite?
- 4 What is its altitude?
- 5 What happens to the period as the altitude increases?

Plenary

EXTRA PRACTICE

- 1 What is meant by the period of a satellite?
- 2 Why does a geostationary satellite stay above one place?
- 3 Which has the shorter period — the ISS or a TV satellite?
- 4 Why does a satellite stay in orbit?
- 5 Describe what happens in Newton's cannon.

Extension

- 1 Why must a TV satellite be geostationary?
- 2 Suggest why weather satellites are often in low orbits.
- 3 Why does a satellite in a low orbit need to travel faster?

Extension

EXTRA PRACTICE

- 1 Suggest why GPS needs many satellites, not just one.
- 2 Why can the ISS be seen crossing the sky in a few minutes?
- 3 Suggest a problem with 36 000 km for a phone call.

Course specification

N5 Physics course specification v4.0, p.8 – Space exploration

- Awareness of the benefits of satellites: GPS, weather forecasting, communications, scientific discovery and space exploration (for example Hubble telescope, ISS).
- Knowledge that geostationary satellites have a period of 24 hours and orbit at an altitude of 36 000 km.
- Knowledge that the period of a satellite in a high altitude orbit is greater than the period of a satellite in a lower altitude orbit.